

3 - Decision Trees

Flipped learning - Please watch the short video below before the start of the lesson



Learning Outcome

To be able to construct, calculate, and evaluate decision trees to support business decision-making.

To Know



- What a decision tree is and explain its purpose in decision-making.
- Construct a simple decision tree diagram showing possible outcomes, probabilities, and expected values.
- Calculations to determine the expected monetary value (EMV) of each option.
- Interpret the figures to recommend a course of action.
- Explain the advantages of using decision trees (e.g. structured decisions, quantifies risk).
- Discuss the limitations (e.g. relies on estimates, doesn't capture qualitative factors).

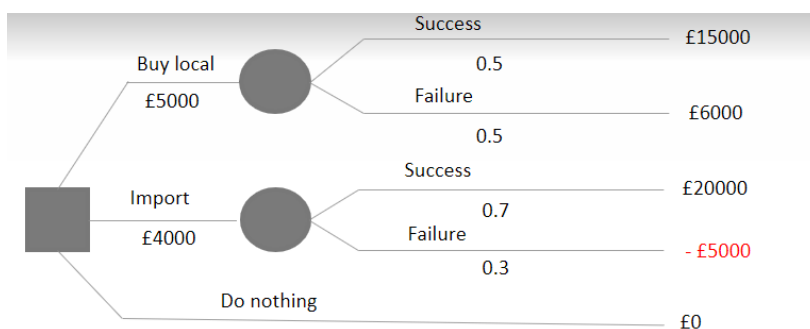
Decision Tree example

Options	Cost	Success		Failure	
		Probability	Financial outcome	Probability	Financial outcome
Buy local	£5000	50%	£15000	50%	£6000
Import	£4000	70%	£20000	30%	- £5000

What does this tell us?

- If the manufacturer chooses to buy their materials from a local supplier it will cost £5000
- If this is successful the financial outcome will be £15000, but there is only a 50% chance of success
- If this is not successful the financial outcome will only be £6000, there is a 50% chance of failure
- If the manufacturer choose to import supplies it will cost £4000
- If this is successful the financial outcome will be £20000, and there is a 70% chance of success
- If not successful the financial outcome will be -£5000, and there a 30% chance of failure

Decision tree



I Do, We Do, You Do

Expected value

First calculate the **expected value** - Multiply the financial outcome by the probability for each chance

Buy local

Success =

Failure =

Expected value =

Import

Success =

Failure =

Expected value =

The expected values are shown in the chance nodes (the circles)

Net gain

Subtract the cost from the expected value

Buy local

Expected value =

Cost =

Net gain =

Import

Expected value =

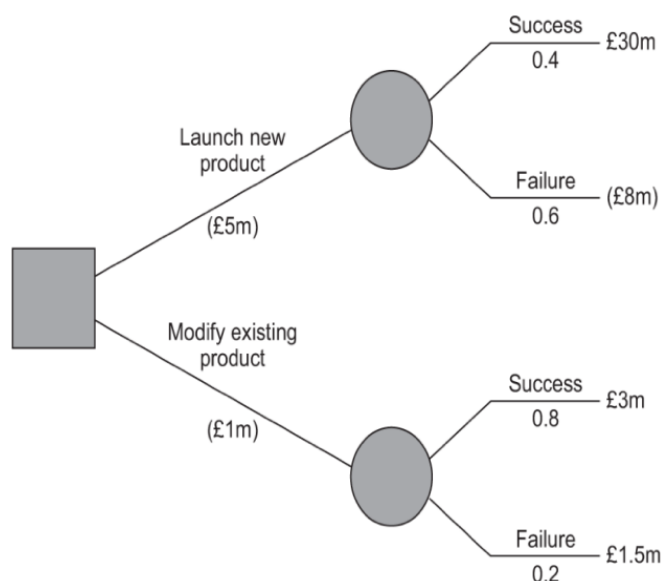
Cost =

Net gain =

The highest net gain is shown in the decision nodes (the square) and the other options crossed off with a single line - **the business would import the supplies based on this decision tree.**



You Do



Using the decision tree model, recommend whether they should launch a new product or modify an existing one?

What are the advantages and disadvantages of using decision trees as a decision making tool?

Practice Question 1



Tata Motors' tiny Nano proves large distraction

A decade on from the high-profile launch of *Tata Motors' Nano*, it sold just 94 units in December 2017. compared with an original prediction that monthly sales would be

POINT However, a disadvantage is ... This is because... IMPLICATION/S Consequently/This means/This will lead to... WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S? Therefore... As a result, ...	
Paragraph 3 POINT One advantage is.... This is because... IMPLICATION/S Consequently/This means/This will lead to... WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE STATED BUSINESS OBJECTIVE? Therefore... As a result, ...	
Paragraph 4 POINT However, a disadvantage is... This is because... IMPLICATION/S Consequently/This means/This will lead to... WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S? Therefore... As a result, ...	
Conclusion: <u>4W's</u> Which option?	Overall, it is recommended that.....

Why?	This is because...
Why reject the other option?	I have rejected the other option because ...
What factors does the success of your chosen option depend upon?	The success of this option will depend on...
	It is recommended that...

Practice Question 2



About Center Parcs

Center Parcs started in 1968 in The Netherlands. A Dutch businessman, Piet Derksen, opened a holiday village with accommodation in the form of tents. As the business grew, tents became luxurious lodges, located in beautiful woodlands with leisure facilities and restaurants.

Center Parcs is a successful business. Holiday villages have been opened in The Netherlands, Belgium, France and the UK. In 2001, Center Parcs took over UK based Oasis Lakeland Holiday Village and renamed it Center Parcs Whinell Forest. 5

Many of the Center Parcs villages closed during the global health crisis but all were reopened by the end of 2021. In 2022, the business announced plans to expand its Longford Forest site in the UK, adding 200 lodges to this very successful village. 10

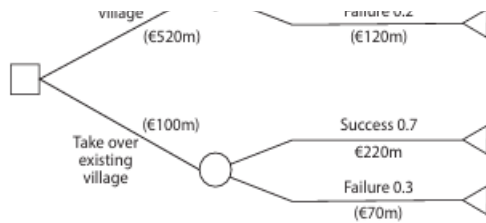
Extract F

Decision tree analysis (DTA)

Center Parcs is considering two options to grow its business. The DTA data and decision tree are shown below:

Option 1 Organic growth: build a new holiday village	Option 2 Inorganic growth: take over an existing holiday village
Cost €520m	Cost €100m
Success estimate 0.8 expected revenue €780m	Success estimate 0.7 expected revenue €220m
Failure estimate 0.2 estimated loss €120m	Failure estimate 0.3 estimated loss €70m





Using the data in Extracts E and F, and decision tree analysis (DTA) calculations, evaluate the extent to which DTA will help *Center Parcs* decide on its growth strategy.

(20)

The objective the business is attempting to achieve is....	
Paragraph 1 POINT IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE STATED BUSINESS OBJECTIVE?	One advantage is.... This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 2 POINT IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Another advantage is ... This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 3 POINT	However, one disadvantage/limitation/point to consider is.... This is because...

IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE STATED BUSINESS OBJECTIVE?	Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 4 POINT IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Another disadvantage/limitation/point to consider is.... This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Conclusion: <u>4W's</u> Which option? Why? What factors does the success of your chosen option depend upon?	Overall, (directly answer the question asked) This is because... Success will depend on... It is recommended that...